TD 2

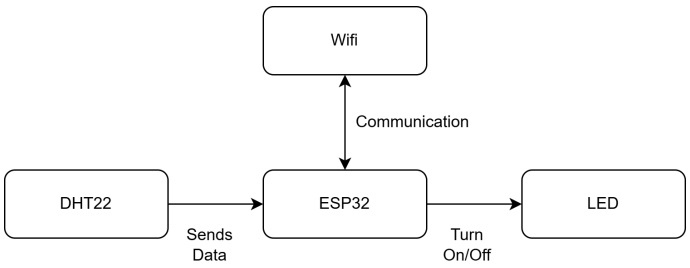
Exercise 1

An IoT system uses an ESP32, a DHT22 temperature/humidity sensor, an LED, and a Wi-Fi connection.

1. Identify the role of each component.
2. Classify each component as input, processing, output, or communication.
3. Draw a block diagram showing how the components interact.

Solution Exercise 1

1. role of each component :
   - **ESP32 :** micro-controller
   - **DHT22 :** sensor
   - **LED :** actuator
   - **Wifi :** connectivity
2. Classification :
   - **ESP32 :** processing
   - **DHT22 :** input
   - **LED :** output
   - **Wifi :** communication
3. Diagram



Exercise 2

Complete the following table:

Sensor	What does it measure or detect?	Output type
DHT22	?	?
LDR	?	?
PIR	?	?
HC-SR04	?	?
MQ-2	?	?

Solution Exercise 2

Sensor	What does it measure or detect?	Output type
DHT22	temperature/humidity	Digital
LDR	light brightness	Analog
PIR	Motion	Digital
HC-SR04	Distance	Digital
MQ-2	Smoke/Gaz	Analog/Digital

Exercise 3

An IoT device uses a 3000 mAh battery. During each minute:

- The sensor consumes 10 mA for 2 seconds.
- The microcontroller consumes 20 mA for 3 seconds.
- The device consumes 0.1 mA during the remaining time.

Assume the three operating periods do not overlap.

1. Calculate the average current consumption.
2. Estimate the battery lifetime in hours and days.
3. Give two ways to increase battery lifetime.

Solution Exercise 3

1. average current consumption:
   - sensors 10mA for 2 seconds
   - microcontroller 20mA for 3 seconds
   - device 0.1mA remaining time = 60-(2+3) = 55 seconds

$$\begin{aligned} \text{avg}_{\text{consumption}} &= \frac{\sum \text{mA}_i \times t_i}{t_{\text{total}}} \\ &= \frac{10 \times 2 + 20 \times 3 + 0.1 \times 55}{60} \\ &= \boxed{1.425\text{mA}} \end{aligned}$$

2. Battery lifetime :

$$\begin{aligned} \text{battery}_{\text{lifetime}} &= \frac{\text{capacity}}{\text{avg}_{\text{consumption}}} \\ &= \frac{3000}{1.425} \\ &= 2105.26\text{h} \\ &= \boxed{87.7\text{days}} \end{aligned}$$

3. Two ways to increase battery lifetime :
   - **Sleep mode :** put in idle mode when not needed to save energy
   - **Lower communication frequency :**
     - **Event Driven :** communicate only when an event is triggered
     - **Periodic :** communicate each t time
     - **Query Driven :** communicate only when a query was sent

Exercise 4

A potentiometer is connected to an ESP32 **ADC** input. Its output voltage varies from 0 to 3.3 V.

1. Is the potentiometer an analog or digital sensor?
2. What is the role of the ADC?
3. For a 12-bit ADC, how many different values can be represented?
4. What digital value approximately corresponds to an input voltage of 1.65 V, assuming an ideal 0-3.3 V range?

Solution Exercise 4

1. The potentiometer is an analog sensor because it uses **ADC** (Analog To Digital Convertor) and it is also mentioned that voltage output varies from 0 to 3.3V
2. The role of the **ADC** to convert analog signal to digital
3. We can represent $2^{12} = 4096$ values in 12-bit
4. digital values for 1.65 V in ideal range 0-3.3V range :

$$\begin{aligned} \text{digital} &= \frac{\text{value}_{\text{volt}}}{\text{max}_{\text{volt}}} \times \text{max}_{\text{digital}} \\ &= \frac{1.65}{3.3} \times (2^{12} - 1) \\ &= \boxed{2047.5} \end{aligned}$$

Exercise 5

Consider three IoT networks:

- **Network A** : All sensors communicate directly with one central gateway.
- **Network B** : Sensors can communicate with neighboring sensors to forward data.
- **Network C** : Two devices communicate directly with each other.

1. Identify the topology used in each network.
2. Which topology offers alternative communication paths?
3. What happens if the central gateway in Network A fails?

Solution Exercise 5

1. Topology used in each network :
   - **Network A** : star
   - **Network B** : Mesh
   - **Network C** : point to point.
2. Mesh topology offers alternative paths
3. If the central gateway in Network A fails the whole system goes down with it (single point of failure)

Exercise 6

A connected door lock uses an ESP32, an RFID reader, RFID cards, and an electronic lock.

1. Identify the hardware components.
2. Identify three software functions required by the system.
3. Explain how the system verifies whether a user is authorized.
4. Why is a relay or suitable driver needed to control some electronic locks?

Solution Exercise 6

1. power supply, drivers, an ESP32, an RFID reader, RFID cards and an electronic lock
2. read RFID $\rightarrow$ Authentication $\rightarrow$ Lock control
3. the reader sends an **ID** to the **ESP32** which will check a list of **IDs** if matched open door else access denied
4. because the microcontroller doesn't need much voltage whereas an electronic lock needs way more therfore using a driver is needed to safely switch the lock's power and protects the microcontroller.

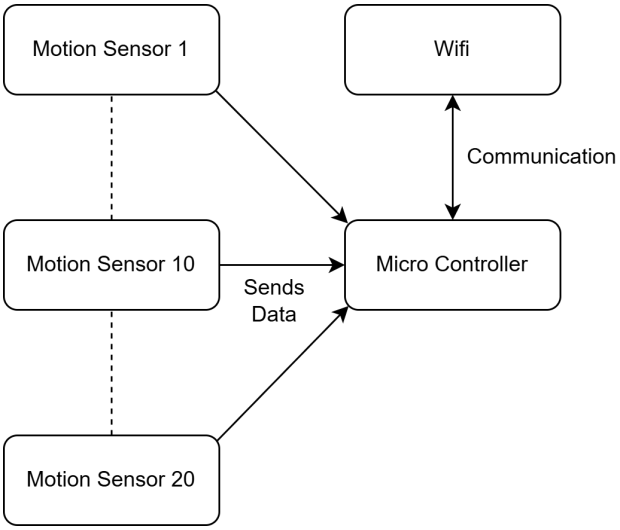
Exercise 7

A parking lot contains 20 parking spaces. The objective is to detect available spaces and display their status on a web dashboard.

1. Which sensors could be used to detect vehicle presence?
2. Identify the processing and communication components.
3. Draw the system architecture.
4. What should the dashboard display if a sensor stops working?
5. Suggest two ways to reduce energy consumption.

Solution Exercise 7

1. Motion sensors
2. processing component is the microcontroller and for communication we will use **Wifi** which is enough to cover parking lot.
3. Diagram :



4. it should display unknown status
5. either lower communication frequency or use sleep mode

Exercise 8

Choose an appropriate actuator or control component for each application and justify your answer.

1. Opening and closing an irrigation water valve.
2. Controlling the speed of a DC motor.
3. Switching a water pump ON and OFF.
4. Moving a robotic arm to a specific angle.
5. Adjusting the brightness of an LED.

Solution Exercise 8

1. Relay or a valve to open or close
2. **DWM** allows for precise, predictable adjustments to linear movement
3. Relay to switch ON or OFF
4. Motor for rotation
5. **PWM** quick switching allows to adjust brightness